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DATABASE MANAGEMENT SYSTEM

ASSIGNMENT

Q1. What is database Management system (DBMS)? Why is it required?

ANSY A database Management system (DBMS) is software that is used to store manage and retrieve data efficiency from a database. It provides a systematic way to organize data so that it can be easily accessed, updated & maintained. A DBMS is required because it ensures data security by restricting unauthorized acess, maintains data consistency & integrity, reduces data redundancy by avoiding duplicate storage and allows quick and easy access to large amount of data. Overall, a DBMS help in managing data efficiently and reliable.

Q2. Why is DBMS better than the traditional file system?

ANSY DBMS is better than the traditional file system because it reduces data redundancy and avoids data inconsistency by storing data in a structural and centralized manner. It provides better data security through authorisation and access control mechanism. A DBMS also supports concurrency, allowing multiple user to access and modify data at sametime without conflict. In addition, it offers backup and recovery features to protect data from loss or system failure which aren't efficiently supported in traditional file system.

Q3. What is data abstraction? explain it's level.

ANSY Data abstraction in DBMS refer to hiding the internal details of data storage & implementation from users to simplify iteractions with database. It has 3 level of abstractions. The physical level explains how data is actually stored in the system. The logical level describes what data is stored and the relationship among the data. The view level present only the required portion of the database to users while hiding unnecessary details.

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Q4. What is the three schema architecture ? Why is it important ?

ANS) The three schema architecture is a DBMS framework that divides the database system into three level: internal, conceptual and external. The internal level describes how data is physically stored, the conceptual level defines the overall logical structure of database and the external level shows different user views of the data. This is architecture is important because it provides data independency, meaning changes at one level do not affect the other levels, and it helps in maintaining a well structured & organised database design.

Q5. What is data independence and why it is needed ?

Ans) Data independence is the ability to change the schema at one level of a database without affecting the schema at other level. It is needed because it provides flexibility in database design and allow easy modification and maintance of the database. With data independence changes in physical storage or logical structure can be made without impacting user applications, making the system more reliable and adaptable.

Q6. What is an Entity Relationship (ER) model ?

ANS) An Entity Relationship (ER) model is a conceptual database model used to represent real world object's called entities, their attributes and the relationship between them in a graphical form. It helps in designing the structure of database by clearly showing how data is connected before it implementing it in a database system.

Q7. What are integrity constraints ? Why are they important in DBMS ?

ANS) Integrity constraint are rules applied to a database to maintain data accuracy and consistency. They ensure that only valid and reliable data is stored in the database. Integrity constraint are important in DBMS. because they prevent invalid data entries and maintain the correctness of relationship between tables. Common example is primary key, foreign key, unique, not null and check constraint.

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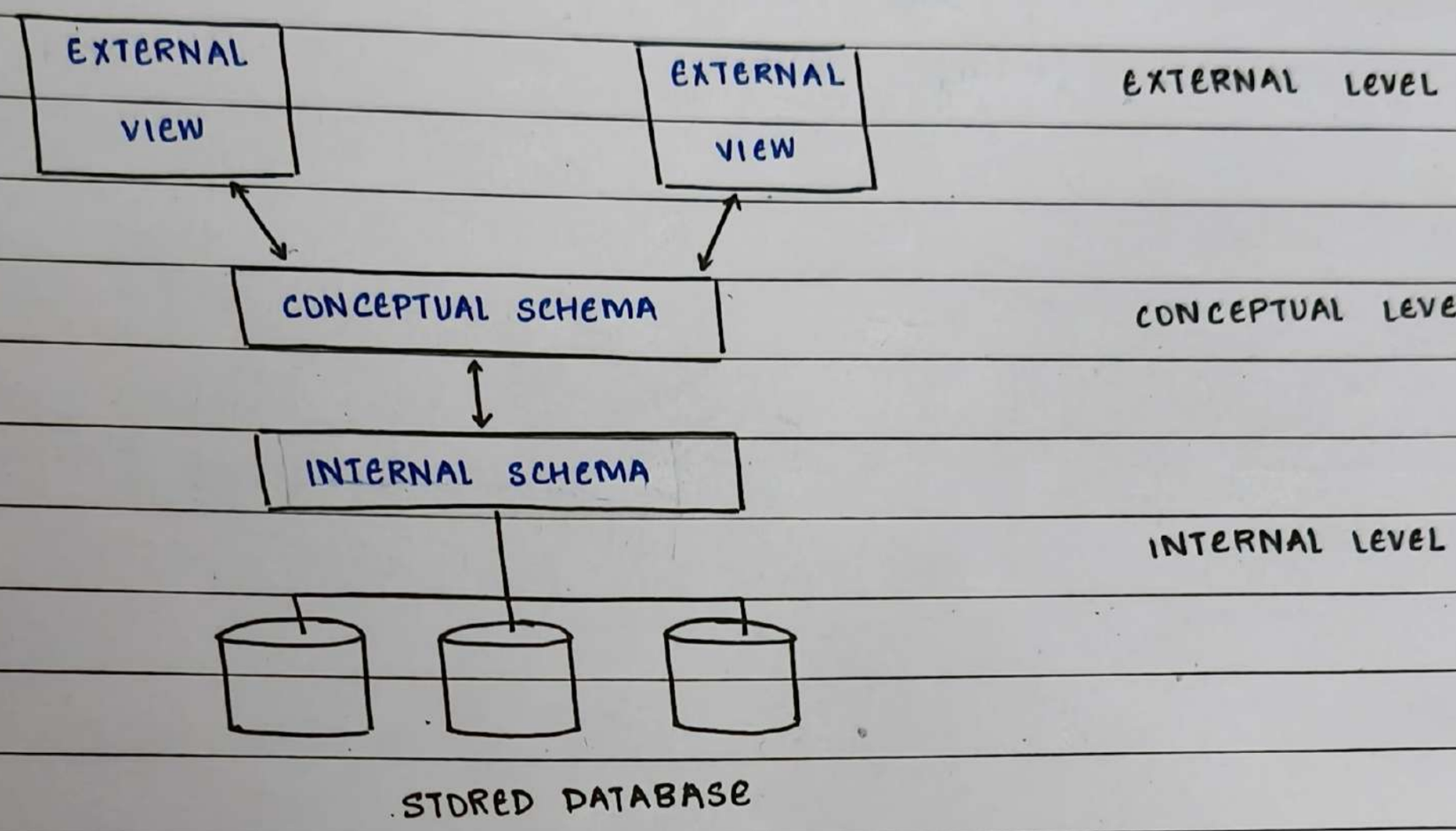
SECTION - B (DESIGN, DRAW & DEFEND)

Q Draw the 3 schema architecture and explain each level.

ANS:



The 3 schema's in a database architecture provide layer of abstraction for data independence :

EXTERNAL SCHEMA :

customised view for specific users or applications, showing only relevant data subset while hiding the rest.

CONCEPTUAL SCHEMA:

Defines the logical structure of the entire database, including entities, relationship and constraint, independent of storage details.

INTERNAL SCHEMA :

specifies physical storage aspect like file organisation, indexes, access path for optimal performance.

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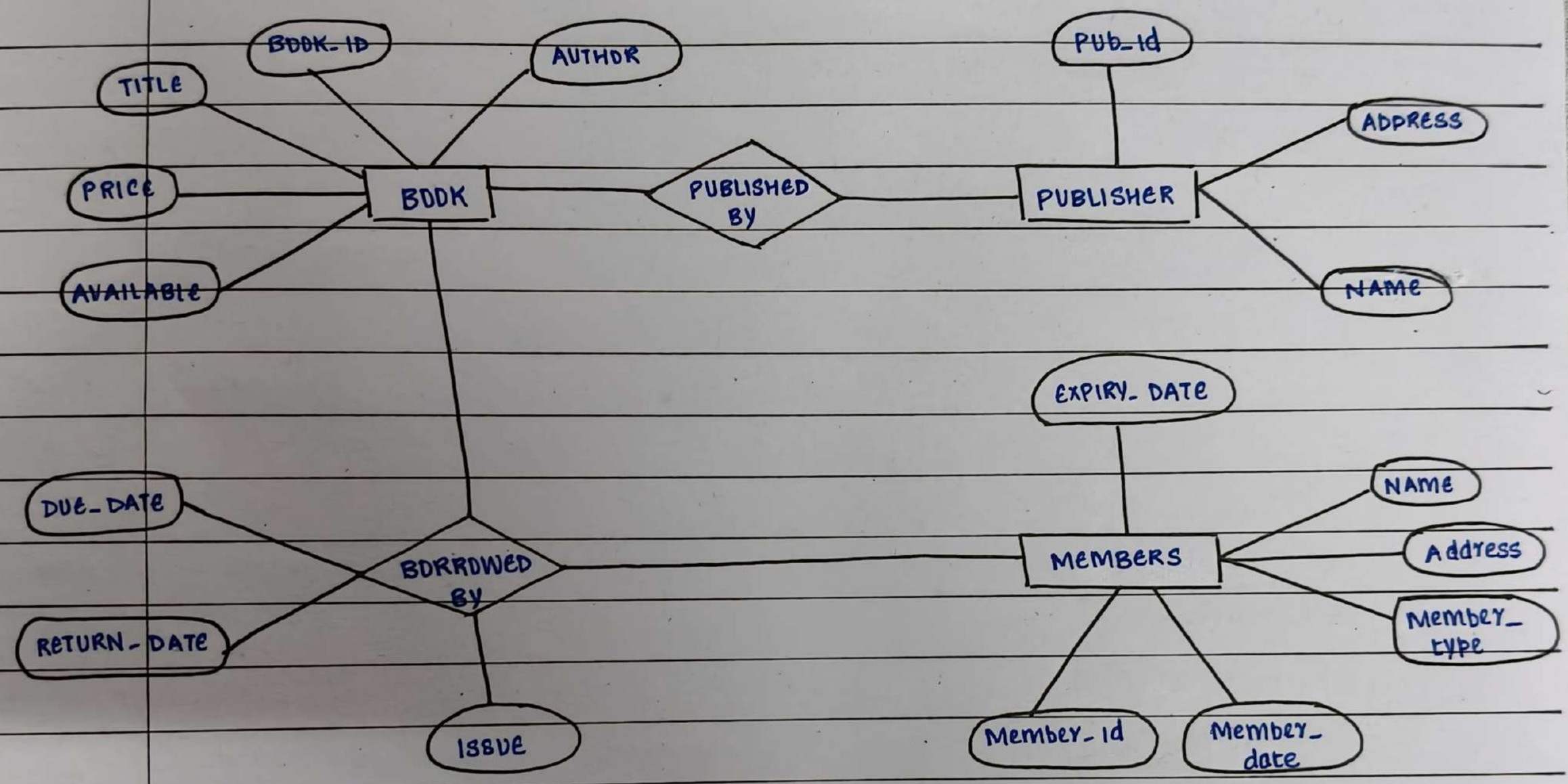
Q. DRAW an ER diagram for a library management system.

ANSY A library management system ER diagram typically includes key entities like BOOK, members, librarian and transaction. with attributes, primary keys and relationship for borrowing / Returning BOOK.



	PRIMARY KEY	KEY ATTRIBUTE
MEMBERS	Member_id	Name, [illegible] address
BOOK	BOOK_id	Title, [illegible], Publishe
PUBLISHER	Pub_id	Name, Address

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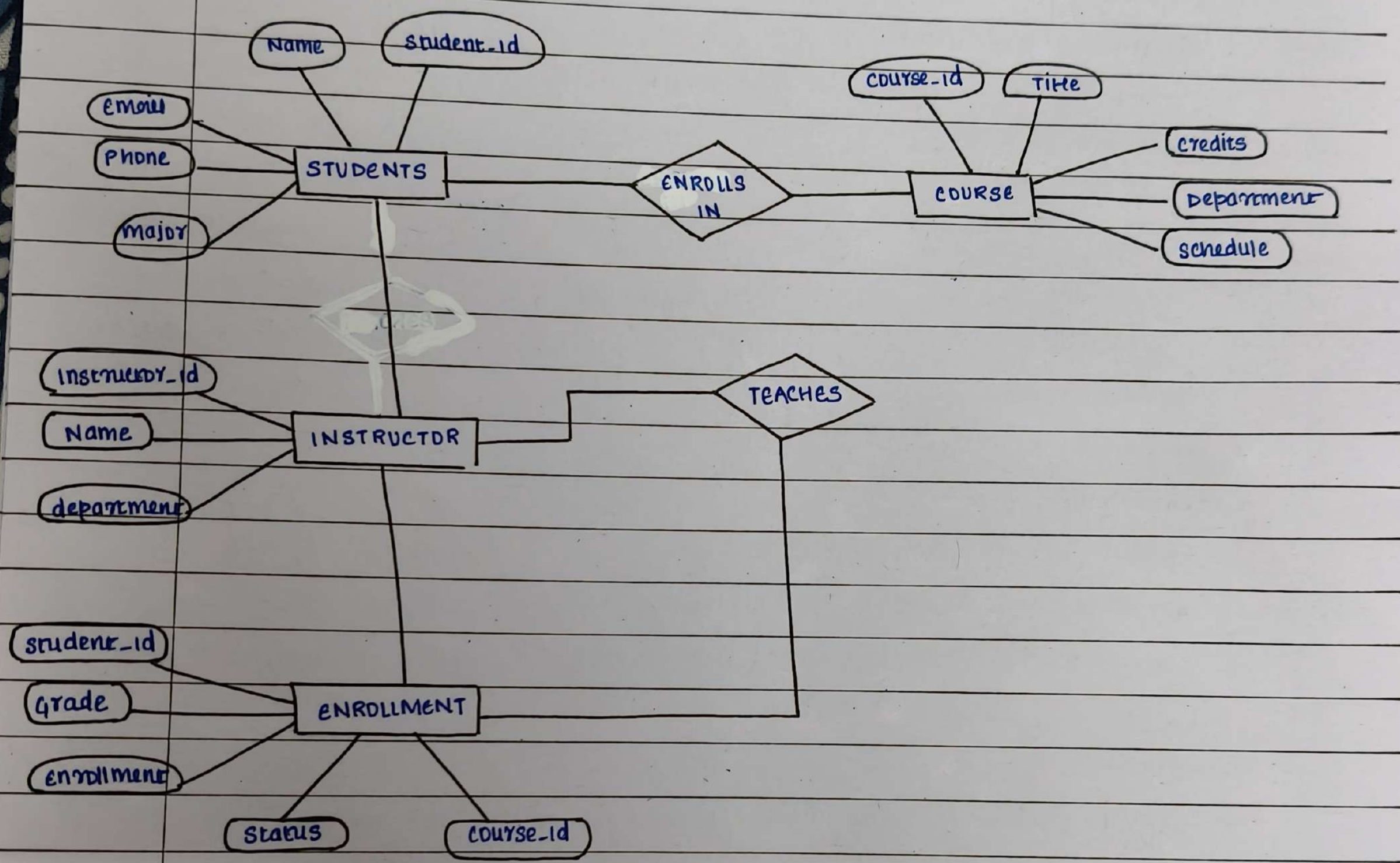
Q Draw an ER diagram for student course Registeration system.

ANSY A Student course Registeration system ER diagram model key entities like student, course, instructor, enrollment to capture Registeration process with proper cardinalities.



	PRIMARY KEY	KEY ATTRIBUTES
STUDENT	student_id	Name, email, Phone_nos...
INSTRUCTOR	instructor_id	Name, department,
COURSE	course_id	title, credits, schedule...
ENROLLMENT		grade, enrolment, status

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SECTION-C (CASE BASED ANALYTICAL QUES)

Q. A hospital wants to store patient, doctor, appointment data. Identify entities, attributes & Relationship.

ANS) ENTITIES & ATTRIBUTES

1. PATIENT :
- Patient_id (P.K.)
- Name
- Age
- Gender
- Address
- Phone_nos.

2. DOCTOR :
- doctor_id (P.K.)
- Name
- specialization
- Phone_nos

3. APPOINTMENT :
- Appointment_id (P.K.)
- Appointment_date
- Appointment_time
- Patient_id (F.K.)
- Doctor_id (F.K.)

RELATIONSHIP

- Patient can book many appointments
- A doctor can have many appointments
- each appointment is associated with 1 patient & doctor.

CONSTRAINTS

Patient_id, doctor_id, Appointment_id must be unique and not null.

Q An online shopping platform stores customer, product and order data. suggest keys and constraints.

ANS) ATTRIBUTES

1. CUSTOMER
- Name
- Email
- Phone
- Address
- customer_id (PK)

2. PRODUCT
- Product_Name
- Price
- stock_quantity
- Product_id (P.K.)

3. ORDER
- Order_date
- Quantity
- Order_id (P.K.)

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PRIMARY KEY

CUSTOMER : customer_id

PRODUCT : product_id

ORDER : order_id

FOREIGN KEY

order table includes :

- customer_id
- links customer to their order

order items :

- orderId
- product_id

INTEGRITY CONSTRAINT

1. NOT NULL : customer_id , product_id, order_id on all referencing table ; essential field like product. Price (>0) , order. totalamount (>=0).

2. unique : customer. email to prevent duplicates

3. CHECK : Product. stock quantity >=0 ; order. instatus ('Pending', 'delivered', 'cancelled').

Q. A university stores students record's. explain how data abstraction explain different users.

ANSY Data abstraction in university's student's records database simplifies access for diverse users through 3 levels : view , logical, physical

1. view level

students see only their grades and enrolled courses via personalised view, hiding other's data and internal structure. faculty access class rosters without seeing physical storage details.

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2. logical level

Admission staff view relationship like student course grade table with entities & constraint, independent of file formats.

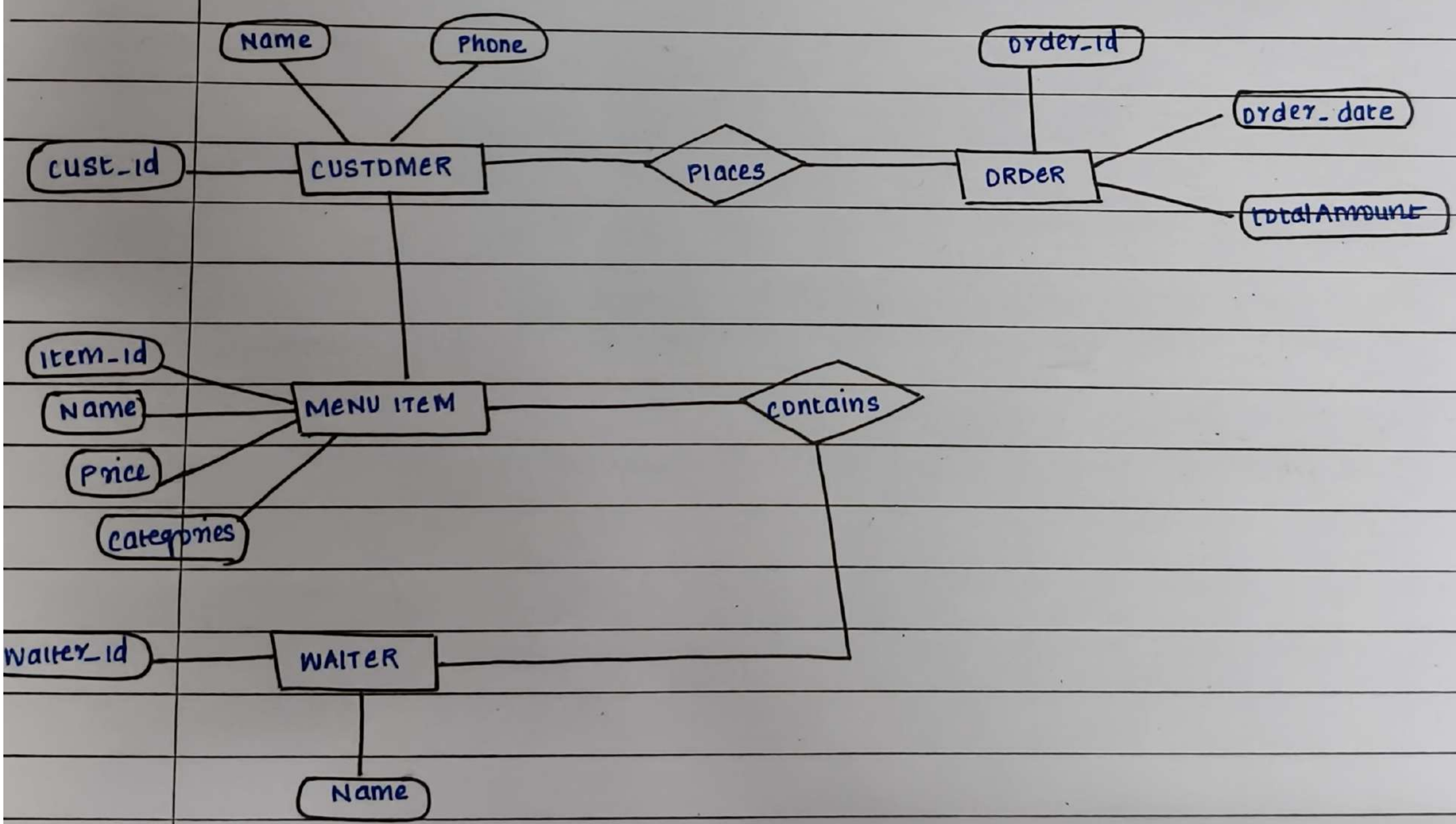
3. Physical level

DBA manages indexes, partitions and storage on servers, changes here don't affect users view due to mapping, ensuring data independence.

SECTION D (one diagram + one insight)

Q Draw a simple ER diagram for any real life system and explain one key design decision.

ANSY A simple ER Diagram for restaurant ordering system:



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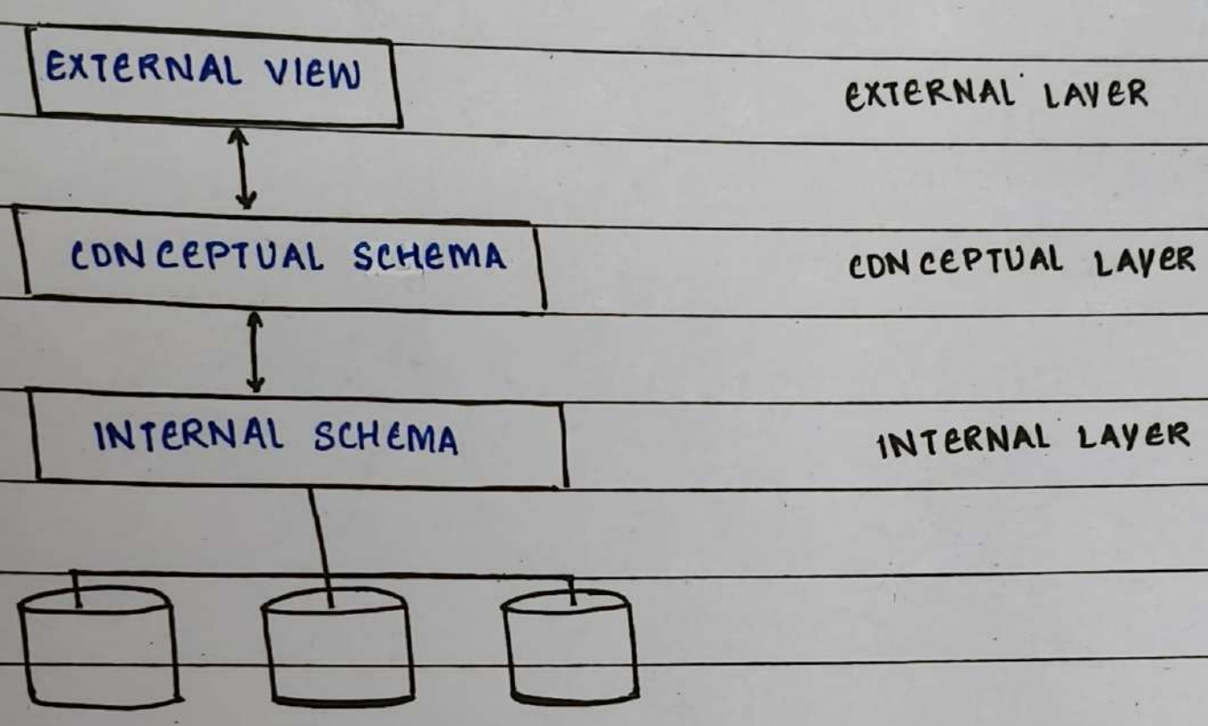
Resolved M:N relationship between order and menu table using composite Primary Key (order_id + item_id).

One order contains multiple menu items and each menu items appears in many orders. direct M:N creates redundancy and update anomilies. this table normalise data, stores quantity per item and supports efficient queries.

Q. Draw 3 schemas architecture and mention one benefit.

ANSY



Key benefit :

data independence :- Physical changes dont affect users view, logical changes dont impact external applications.

Q. Draw an entity with attributes and identify it's Primary Key.

STUDENT

PK student_id

Name

Age

Gender

Email

Phone_nos.

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PRIMARY KEY

- student_id
- uniquely identifies each students
- it cannot be null

ATTRIBUTES

- student_id - unique identification
- Name - Name of student
- Age - Age of student
- Gender - Gender of student
- Email - Email address
- Phone_nos - contact number

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